



# How Artificial Intelligence Has Changed Search Engine Optimisation Practices

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## Abstract

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In this Thesis, the ways in which Artificial Intelligence (AI) has revolutionised the way Search Engine Optimisation (SEO) is conducted will be explored. Drawing on extensive research in the literature and a case study of the widely used AI tools, Semrush and Ahrefs, the paper explores the role of machine learning, natural language processing, and generative AI in the transformation of the content creation, key search process, and ranking system of search engines. The analysis model is based on three theoretical frameworks: Technology Acceptance Model (TAM), Resource-Based View (RBV) and Stimulus-Organism-Response (S-O-R) model. It reveals that AI has completely revolutionised SEO, from a more rule-based and technical practice to an intelligence-based one. The tools' efficiency, level of analysis and strategic decision making are immensely improved, but it requires user experience and strategic integration, rather than simple adoption. The main constraints mentioned are challenges, such as a lack of knowledge of algorithms and privacy of data, and the risk of homogenising content. Practical recommendations can be used by businesses that intend to ensure that the use of AI in their SEO strategies is sustainable and ethical at the end of the study.

**Keywords:** Artificial Intelligence, Search Engine Optimisation, Machine Learning, Natural Language Processing, Digital Marketing, Generative AI, SEO Tools

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## List of Abbreviations

|                  |                                                                                                  |
|------------------|--------------------------------------------------------------------------------------------------|
| <b>AI</b>        | Artificial Intelligence                                                                          |
| <b>BERT</b>      | Bidirectional Encoder Representations from Transformers - a Google NLP algorithm (2019)          |
| <b>E-A-T</b>     | Expertise, Authoritativeness, and Trustworthiness - Google content quality framework             |
| <b>E-E-A-T</b>   | Experience, Expertise, Authoritativeness, and Trustworthiness - updated Google quality framework |
| <b>GDPR</b>      | General Data Protection Regulation - EU data privacy law                                         |
| <b>LLM</b>       | Large Language Model - a deep learning model trained on large text corpora (e.g. GPT-4)          |
| <b>ML</b>        | Machine Learning - a subset of AI enabling systems to learn from data                            |
| <b>MUM</b>       | Multitask Unified Model - Google's multimodal AI algorithm (2021)                                |
| <b>NLP</b>       | Natural Language Processing - AI branch enabling computers to understand human language          |
| <b>RBV</b>       | Resource-Based View - strategic management theory on competitive advantage through resources     |
| <b>RankBrain</b> | Google's first deep learning search algorithm, was deployed in 2015                              |
| <b>SEO</b>       | Search Engine Optimisation - the practice of improving website visibility in search engines      |
| <b>S-O-R</b>     | Stimulus-Organism-Response - a consumer behaviour theory applied to content and user reaction    |
| <b>TAM</b>       | Technology Acceptance Model - a framework for understanding user adoption of new technology      |

# **1 Introduction**

## **1.1 Background of the Study**

The background of the study gives a summary of the proposed study. One of the most important aspects of online marketing has been Search Engine Optimisation (SEO), which enables businesses to have more presence on the internet and have organic traffic through search engines (Tatikonda et al., 2024). In the past, the focus of SEO strategies was to make the most possible use of keywords, build backlinks and control the technicality of websites in order to acquire a more favourable position in search results. However, with the Artificial Intelligence (AI) at a fast pace, the character of SEO shifted significantly, where the emphasis was shifted on intelligent and data-driven methods of SEO, rather than on rules (Davenport et al., 2020; Dwivedi et al., 2021). Artificial Intelligence may be explained as the development of technologies that could implement the functions which human intelligence would perform in general, such as learning, reasoning, and problem-solving (Russell and Norvig, 2021). In the context of online marketing, AI technologies (including machine learning), natural language processing (NLP), and predictive analytics have revolutionised the process of search engines both in terms of interpretation and ranking (Sharma et al., 2025). More search engines, such as Google, increasingly rely on algorithms that are driven by AI to understand the purpose of the user, relevance of the content, and give personalised search results (Huang and Rust, 2021). This has transformed the main principles of SEO to a large extent. SEO is now more focused on high-quality content, user experience, and semantic relevance (Vashishth et al., 2025) rather than on the density of keywords and technical characteristics. Consequently, businesses need to transform their search engine strategies to suit these evolving algorithms. Furthermore, as generative AI is on the increase, the process of creating and optimising content has been taken to a new level. It can be applied to AI to produce quality content, offer keywords and process a massive amount of data to identify patterns and trends. These solutions help to make the work of marketers more efficient and help them make more efficient

decisions with the help of data (Verma et al., 2021). In the meantime, they provide certain concerns about the originality of the content, its ethical use, and the potential over-reliance on automated processes (Feuerriegel et al., 2024). Moreover, AI-powered search engine optimisation tools like Semrush and Ahrefs have become a crucial part of contemporary digital marketing. The tools utilise AI to give an insight into the performance of keywords, competitor strategies, and opportunities to optimise websites. The latter allows marketers to create more targeted and effective SEO strategies through real-time analysis of large datasets (Chaffey and Ellis-Chadwick, 2022). With these changes, it is clear that Artificial Intelligence is radically changing the way SEO is practised. This transformation is vital to understand in the context of the business being interested in staying competitive in a more digital and AI-driven world.



## 1.2 Research Problem

However, with the increasing adoption of Artificial Intelligence in search engine optimisation, businesses and digital marketers do not have a clear picture of how AI-powered technologies impact SEO practices and performance (Hasan, 2025). The old techniques of SEO that were previously useful in enhancing search rankings are no longer relevant in a world of smart algorithms and the use of data to make decisions (Berman and Katona, 2013). The complexity of AI systems employed by search engines is one of the major issues. These systems keep changing, and practitioners cannot have full insight into the way ranking algorithms work and how SEO strategies need to change in line with them. Consequently, numerous organisations cannot keep up with the rapid technological changes and effectively optimise their digital representation (Dwivedi et al., 2021). Additionally, although AI-driven tools can have great advantages, such as automation, efficiency, and better analysis, there is not much information available as to how AI-based tools can be utilised effectively and ethically. The problem of content authenticity, algorithmic bias, and excessive reliance on automation leaves marketers in uncertainty and poses valuable questions regarding the sustainability of AI-driven SEO practices in the long term (Mowshowitz and Kawaguchi, 2021). The other important issue is the disconnect between theoretical and practical learning. Even though the academic literature has shown that AI has the potential to revolutionise the marketing practice, most companies do not have the practical skills that would enable them to integrate AI tools into their marketing strategies (SEO). This lack of connection restricts the potential of organisations to maximise the advantages of AI technologies (Kietzmann et al., 2023). In such a way, the research problem that is the focal in this research is the fact that the ways Artificial Intelligence is changing the practices of SEO and how businesses could adapt to those changes successfully to increase their visibility and performance online is not known in-depth.

## **1.3 Research Aim and Objectives**

### **1.3.1 Research Aim**

The main objective of this research is to discuss how AI has changed the way people optimise their search engines and to find effective strategies to use AI in search engine optimisation.

### **1.3.2 Research Objectives**

- To examine the purpose of Artificial Intelligence in search engines and online marketing.
- To investigate the impact of AI technologies on the main SEO activities, such as the content creation, the work of key search engines, and search engine optimisation.
- To investigate the usefulness of AI-based SEO tools like Semrush and Ahrefs.
- To determine the advantages and issues related to the use of AI in SEO.
- To give feasible suggestions that businesses can use to adjust their SEO techniques in the era of Artificial Intelligence.

## **1.4 Research Questions**

- What impact has Artificial Intelligence had on search engine optimisation?
- How do the AI technologies affect the content production, research of keywords and ranking?

- What is the effectiveness of AI-powered SEO tools to enhance digital marketing performance?
- What are the main issues and drawbacks related to using AI in SEO?
- What are the ways that businesses can successfully implement AI-powered SEO?

### **1.5 Significance of the Study**

This work has a great scholarly and practical value. Academically, it is adding to the existing literature on Artificial Intelligence and digital marketing by discussing the overlap of AI and SEO. It offers a detailed insight into the effects of technological progress in changing the conventional marketing processes and suggests the most promising areas of future studies (Cao and Tian, 2023). In practical terms, these study findings can be used by businesses, digital marketers and SEO professionals in a very great way. The study also gives valuable details about the use of AI tools and technologies in online marketing to optimise search engine rankings, enhance organic traffic, and enhance search engine tactics. Having a clue about how AI will affect SEO, companies will be in a better position to make decisions and develop more effective marketing strategies (Jarvinen and Karjaluoto, 2020). Additionally, this paper has suggestions on how AI can be used ethically and responsibly in SEO. As AI advances, it is important that companies adopt practices that ensure that they are transparent, fair, and authentic. The paper illuminates such considerations and helps in the development of sustainable and ethical SEO strategies (Feuerriegel et al., 2024). Also, the work can be of use to students and researchers interested in the areas of Artificial Intelligence, digital marketing, and optimisation of search engines. It also provides a clear image of the current trends and developments in the future, hence warranting future academic research in the field (Verma et al., 2021).

## **1.6 Scope of the Study**

The purpose of this paper is to analyse the impact of Artificial Intelligence on the search engine optimisation processes. It gets into the applications of AI technologies that comprise machine learning, natural language processing and generative AI and their effects on the primary SEO functions, namely the development of content, creation of keywords, and search engine ranking. The research is a study based on the research and is implemented in a qualitative manner with a substantial literature review and examination of the case of the use of AI-driven SEO tools, such as Semrush and Ahrefs. These tools are analysed with the purpose of appreciating the way they are applied in the real world and how effective they are in the existing SEO practices. The study has, however, a couple of limitations. It has nothing to do with primary data collection, empirical tests and software development. Instead, it is based on secondary data only and the analysis of existing literature and tools. Also, the results can be constrained by the swift development of AI technologies that can cause changes in SEO practices in the future.

## **1.7 Structure of the Thesis**

The thesis is divided into five chapters, which cover a particular aspect of the research. Chapter 1: Chapter 1 gives an introduction to the study, which includes the background, research problem, aim, objectives, research questions, significance and scope of the research. Chapter 2 is a literature review on the topic of Artificial Intelligence and search engine optimisation. It analyses the available theories, models, and empirical research on the topic of AI-driven marketing and SEO practices. Chapter 3 describes the research methodology of the present study. It describes the research design, data sources, data collection methods and analytical techniques used to attain the objectives of the research. Chapter 4 will be the analysis and discussion of findings. It assesses how AI will impact SEO practice using the literature review and case studies, and critically interprets the findings. Chapter 5: Chapter 5 will end the research by summarising the main findings, discussing implications and giving practical recommendations on how businesses should act, as well as recommendations on future research directions.

## **2 Literature Review**

### **2.1 Introduction**

The chapter is a critical analysis of the literature regarding the topic of Artificial Intelligence (AI) and its impact on search engine optimisation (SEO). The chapter does not just provide a summary of earlier studies but evaluates the antagonistic arguments, identifies theoretical premises, and indicates the flaws of the existing research. The increased use of AI in digital marketing has transferred the rule-of-thumb in SEO to the dynamic, data-driven processes (Sarfandi, 2025). However, despite the fact that the literature on this subject is slowly growing, it remains more or less disjointed, with no convergence between theoretical understanding and application. The chapter, therefore, is a critical review of the key themes, including AI in marketing, how SEO has evolved, AI-assisted content strategies and AI tool use, and identifies gaps that warrant the present study.

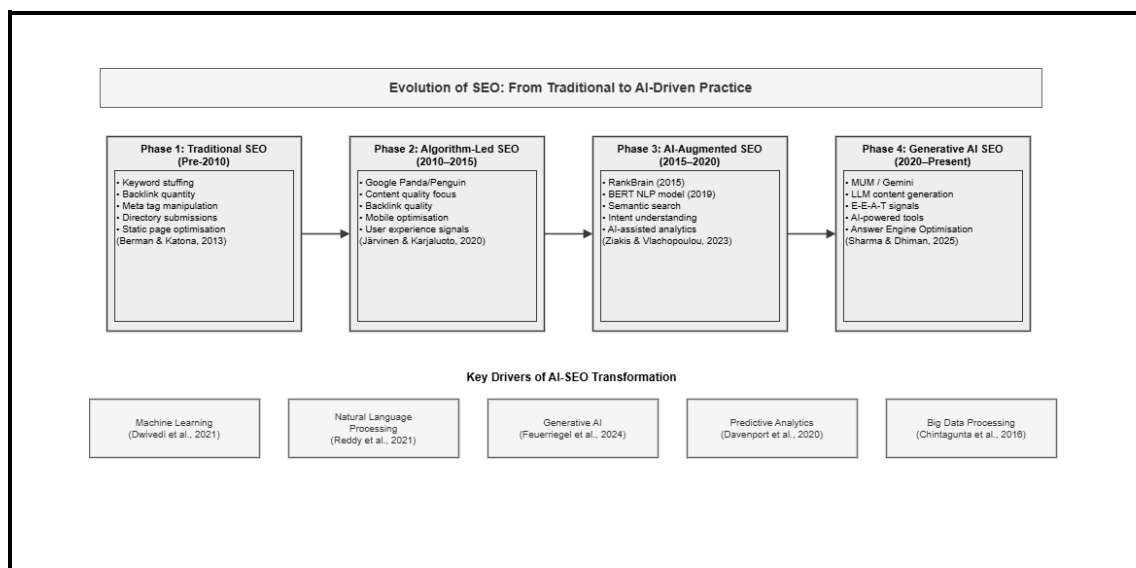
### **2.2 Artificial Intelligence in Marketing**

AI has turned into one of the major marketing innovations, which has enabled corporations to work with a high level of data volume, make decisions automatically and supply personal experiences. According to Russell and Norvig (2021), AI is basically the systems that can perform functions such as learning, reasoning, and problem solving that are typically associated with human intelligence. Marketing AI finds many applications within customer analytics, recommendation systems and predictive modelling.

Davenport et al. (2020) believe that AI is revolutionising marketing by improving efficiency and making decisions in real-time. Equally, Huang and Rust (2021) suggest that AI is transforming service delivery as it moves towards human-centric to hybrid human-AI interactions. Although these studies indicate the transformative nature of AI, they usually lean towards an optimistic viewpoint that does not focus on practical limitations.

Conversely, Verma et al. (2021) highlight that the adoption of AI among organisations is not uniform based on cost, complexity of technology and expertise. This implies that the advantages of AI are not equally available to all companies, especially those that are small. In addition, Chintagunta et al. (2021) also point out the difficulties related to the handling and interpretation of large datasets, which can be misleading unless it is done correctly.

Theoretically, the Technology Acceptance Model (TAM) is a helpful framework to interpret the adoption of AI. TAM argues that ease of use and perceived usefulness affect the use of new technologies (Davis, 1989). Applied to AI marketing tools, this means that organisations will be more likely to use AI when they feel that it is useful and easy to use. Nevertheless, literature seldom utilises TAM as it is in the context of AI-driven SEO, which implies that it does not incorporate it theoretically.



*Figure 2.1: Evolution of SEO from Traditional Keyword-Based Practice to AI-Driven Optimisation (adapted from Berman & Katona, 2013; Ziakis & Vlachopoulou, 2023)*

### **2.3 Evolution of Search Engine Optimisation**

SEO has become a strategic discipline that emphasises user experience and content relevance instead of the technical practice that is based on the manipulation of search engine algorithms (Guelialia and Bouziane, 2024). The initial methods of SEO were based on keyword stuffing, the number of backlinks and technical optimisation. These strategies, however, lost their effectiveness as search engines created more advanced algorithms.

Berman and Katona (2013) state that the current state of SEO focuses on user intent and content quality and not on technical manipulation. To a large extent, this movement is fueled by AI that allows search engines to consider the content more efficiently. Likewise, Järvinen and Karjaluo (2020) note the increased significance of data analytics in the measurement of digital marketing performance.

Regardless of this development, there is still literature that considers SEO as a fixed process, which does not consider the dynamic nature of SEO. This is one of its weaknesses because SEO is being affected more by AI-driven algorithms that are ever-changing, depending on human behaviour.

The Resource-Based View (RBV) has been applied to explain SEO as a strategic capability in a helpful theoretical framework (Kero and Bogale, 2023). RBV supports that competitive advantage is achieved in the form of valuable and rare resources in the organisation. In this regard, state-of-the-art SEO tools with AI support can be seen as a strategic asset that improves the performance of the firms. Nonetheless, there is a gap in the literature as empirical studies seldom relate RBV to SEO practices.



## **2.4 Role of Artificial Intelligence in SEO**

The use of Artificial Intelligence has fundamentally changed the way search engines work, as they can now better process information and provide more relevant results (Ziakis and Vlachopoulou, 2023). Machine learning and natural language processing are AI technologies that help search engines to comprehend user intent and assess the quality of content (Reddy et al., 2021).

The use of AI has transformed the optimisation of SEO to be semantic, meaning that it is focused on meaning, not exact keywords (Ziakis and Vlachopoulou, 2023). This shift imposes on businesses the need to generate quality content, contextual in nature.

Dwivedi et al. (2021) emphasise that the algorithms that rely on AI are constantly getting better with user interactions and enhancing search accuracy and personalisation. The fact that there is no transparency in these algorithms is, however, a big challenge. In the article by Mowshowitz and Kawaguchi (2021), the authors suggest that the level of algorithmic opaqueness can complicate the issue of comprehending the way the search results are ranked, leaving practitioners uncertain.

The theory of Information Asymmetry can be used to study this problem, as we have one party (search engines) more informed than the other (marketers). Such an imbalance restrains the optimisation of the strategies of businesses.

Moreover, although AI improves efficiency, it diminishes control among marketers, with the decisions being made more by automated systems. This raises an accountability and strategic decision-making question.

## **2.5 AI in Content Creation and Optimisation**

One of the areas that has been greatly affected by SEO is content creation. Generative AI tools have the potential to generate content, recommend keywords and optimise content to search engines.

According to Verma et al. (2021), AI can enhance productivity by automating repetitive work and allowing it to generate content at a high scale. Feuerriegel et al. (2024) also emphasise the potential of generative AI to revolutionise content strategies, as large language models can generate contextually consistent, topically relevant content in large quantities.

But there are great challenges to these benefits. The quality and originality of AI-generated content are among the major concerns. Although AI can generate grammatically correct content, it is usually uninspired and two-dimensional. The authors state that AI-generated content can cause homogenization, making the differentiation among competitors less prominent (Cao and Tian, 2023).

Also, moral issues, like plagiarism, misinformation, and transparency, need to be addressed. Excessive dependence on AI tools can also decrease human creativity and critical thinking.

Theoretically, the Stimulus-Organism-Response (S-O-R) framework can be used to comprehend the impact of the AI-generated content on user behaviour. The perception of the user (organism) is influenced by the AI-generated content (stimulus) and, in turn, determines engagement and decision-making (response) (Biswas et al., 2025). However, this theoretical application, in general, lacks in the literature.

## 2.6 AI-Powered SEO Tools

SEO solutions Artificial intelligence has entered the digital marketing market with the Semrush and Ahrefs AI-based tools, which provide a more sophisticated way to analyse keywords, competitors, backlinks, and optimise content (Sivahari, 2025). They use machine learning models and data processing at scale to provide actionable insights so that marketers can make more strategic decisions. Such tools may positively impact the decision-making process because Chaffey and Ellis-Chadwick (2022) suggest that they could turn unstructured data into patterns that can make the implementation of evidence-based marketing strategies possible. Similarly, Kietzmann et al. (2023) emphasise that AI-based tools can be considered successful in personalisation and customer engagement due to their capability to provide a more targeted way of delivering content and optimising it.

Although these benefits are evident, the literature available tends to overestimate the usefulness of AI-based SEO tools by presupposing a linear correlation between the usage of a particular tool and the performance enhancement. This supposition does not take into account the key factor of user experience in decoding and implementing the insights produced by these tools. Practically, the utility of AI-based tools depends on the skills of the user to critically analyse the data, place the results in a context and apply them to the overall strategic plans (Ayoub and Sopuru, 2026).

Also, the literature is more inclined towards a technologically deterministic view, which suggests that the availability of hi-tech tools naturally provides a competitive advantage. This perception contradicts the Resource-Based View (RBV) that competitive advantage does not only come about as a result of having resources but also as a result of using the resources well. In this respect, AI-based SEO tools may only be added to the competitive advantage in case of integration with such organisational resources as analytical capabilities, strategic thinking, and domain knowledge (Hasan, 2025).

The other weakness in the literature is that there is no empirical validation of the actual-world effect of AI-powered SEO tools. Although most studies affirm the theoretical advantages, there is a lack of research to measure their usefulness in various industries and organisational settings. Moreover, the majority of the literature does not take into account the possible dangers that excessive dependence on automated tools may entail, including less human control and more susceptibility to the alterations of algorithms.

In addition, AI-based applications are usually black boxes, which give advice without showing the processes that are going on (Shafik, 2026). This non-explainability brings about questions of accountability and trust, especially in cases where automated insights are being used to make decisions.

## **2.7 Benefits and Challenges of AI in SEO**

The incorporation of Artificial Intelligence into SEO can be seen as having a number of important advantages, which fundamentally transform the methods of developing and executing digital marketing strategies (Guelialia and Bouziane, 2024). Among the main benefits, it is to enhance efficiency as AI can automate processes such as the analysis of keywords, optimisation of content, and tracking performance. This would enable the marketers to concentrate on more strategic-level operations and not on manual activities. Also, AI is more accurate as it processes large volumes of data and finds patterns that could be difficult to observe with conventional approaches.

Jarvinen and Karjaluoto (2020) emphasise the role of data analytics in enhancing marketing performance and state that basing decisions on data results in more successful strategies. Similarly, Kietzmann et al. (2023) observe that AI enables the effortless personalisation, therefore marketers can distribute personalised content that appeals to the preferences and behaviour of the user. The result of such a sense of personalisation, besides improving the user experience, enhances engagement and conversion rates. However, despite these benefits, there are some issues related to AI usage in SEO. One of the most critical problems is a lack of transparency in the AI algorithms. As Mowshowitz and Kawaguchi (2021) claim, algorithmic opacities complicate the comprehension of the process of decision-making and marketers fall short. This transparency robs organisations of the authority to holistically manage and optimise their SEO practices. Another critical issue that leads to bias or unfair outputs is the algorithm's bias. Historical data is used to train AI systems, and it may be biased.

The issue of data privacy is also a significant concern, since the application of AI-based SEO is based on the idea that the data provided by the user is crucial to the process of creating insights and content personalisation (Vajrobol et al., 2024). Moreover, the fact that technology changes rapidly is a major challenge to businesses, and it needs to constantly learn and develop capabilities (Challoumis, 2024).

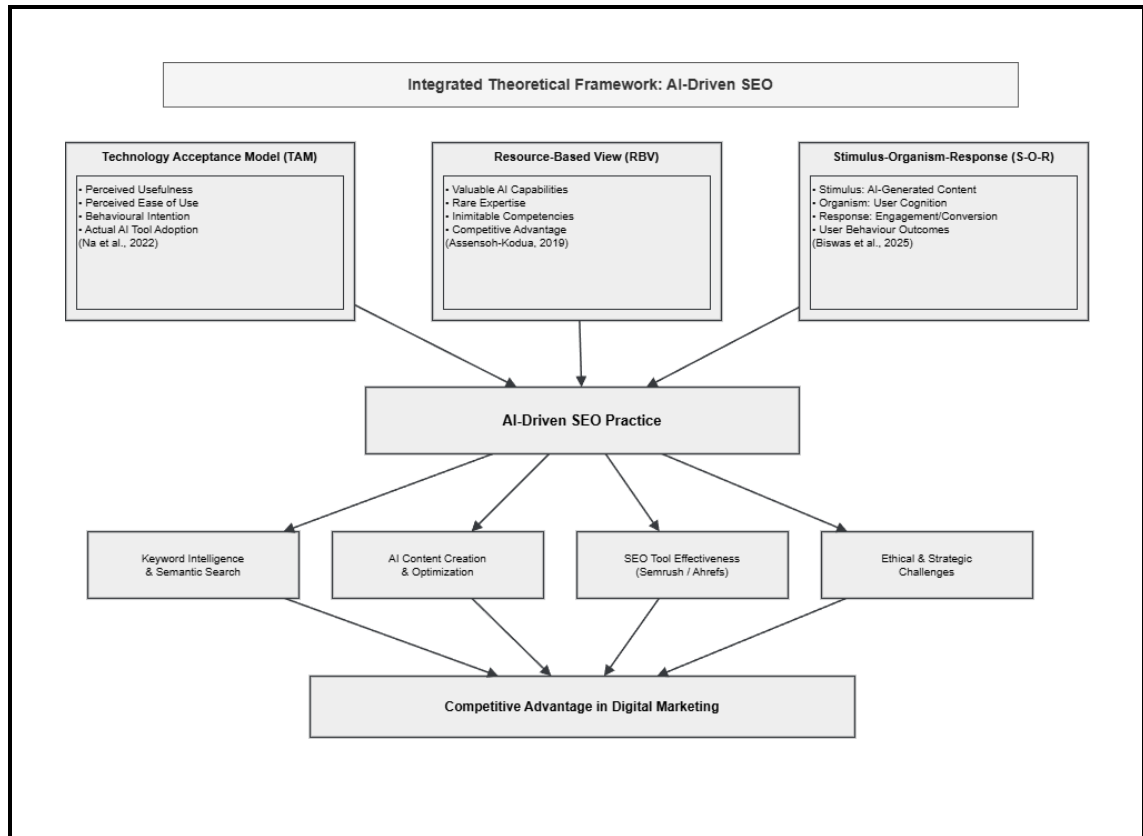
## **2.8 Theoretical Integration**

The lack of theoretical integration is the defining feature of the existing literature on AI and SEO. The current study attempts to overcome this limitation through the incorporation of three important theories, namely; the Technology Acceptance Model (TAM), the Resource-Based View (RBV), and the Stimulus-Organism-Response (S-O-R) model.

The Technology Acceptance Model proposes a model that can be used to understand the adoption of AI technologies by organisations (Na et al., 2022). Perceived ease of use and usefulness determine the need to adopt according to TAM. Applying this to the case of AI-driven SEO engines, this implies that organisations tend to embrace them more when they find them useful and simple to use.

The Resource-Based View is a strategic approach which assumes that competitive advantage is based on valuable, rare, and inimitable resources (Zvarimwa and Zimuto, 2022). AI capabilities may be viewed as a strategic resource which improves the performance of the organisation. Nevertheless, RBV also emphasises that resources are not enough; organisations should also have the capabilities to use them (Assensoh-Kodua, 2019).

The Stimulus-Organism-Response model offers information on how users behave and how external stimuli (e.g., AI-generated content) affect inner cognitive processes and behavioural responses (Biswas et al., 2025). Combining these theoretical views, this research is able to offer a more thorough insight into AI-based SEO.



*Figure 2.2: Integrated Theoretical Framework - TAM, RBV, and S-O-R Applied to AI-Driven SEO*

## **2.9 Research Gap**

Even though the field of Artificial Intelligence and digital marketing is becoming increasingly well-researched, there are still a number of gaps that need to be addressed. First, most of the available studies concentrate on AI in marketing in general, and few studies comment on its particular implications on SEO (Ziakos and Vlachopoulou, 2023). Second, empirical studies assessing the usefulness of AI-based SEO tools in the real world are lacking (Sharma and Dhiman, 2025). Third, theoretical models like TAM, RBV, and S-O-R are seldom combined in the literature on AI-based SEO. Also, the literature is usually overly positive about AI and does not consider the issues of bias in algorithms and ethics (Elkhatibi and Boussedra, 2025). Lastly, there is a lack of academic-research-practical fit. This paper fills these gaps by offering a critical and integrated approach to analysing AI-based SEO.

## **2.10 Chapter Summary**

This chapter critically analysed the literature on AI and SEO, outlining important themes, theoretical views, and gaps in the literature. It proved that AI has turned SEO into a data-driven and user-oriented field. Nonetheless, issues of transparency, ethics, and implementation are still relevant. The gaps in the literature have also been listed in the chapter, such as the absence of theoretical integration and practical insights, and this is the reason why such a study is important.



### **3 Research Methodology**

#### **3.1 Introduction**

The chapter introduces the methodological framework used to discuss the changes brought about by Artificial Intelligence (AI) to search engine optimisation (SEO) practices. It describes the research philosophy, method, design, data sources and methods of analysis applied in the study. Also, the chapter critically justifies such methodological decisions against the objectives of the research, and comments on ethical considerations and limitations. The purpose is to provide the methodological rigour, transparency, and correspondence to the theoretical frameworks that were provided in Chapter 2.

#### **3.2 Research Philosophy**

The positivist research philosophy underlies this paper as it focuses on objectivity, empirical data, and the methodical study of observable facts. Positivism presupposes that knowledge is attainable by logical reasoning and analysing available information, which is why it is especially appropriate in studies that strive to determine patterns and connections within the already known research (Saunders et al., 2019).

Positivism should be adopted in this research because the research aims to measure the effects of AI on SEO activities by conducting an analysis of secondary data and literature. The AI-based SEO is primarily influenced by quantifiable and monitorable aspects of changes in the algorithm, the performance of the content and user-engagement indicators, which correspond to the positivist focus on objective analysis (Dwivedi et al., 2021). Nonetheless, it is worth noting that AI systems tend to be black boxes, which restricts the entire transparency and makes it difficult to assume the impossibility of complete objectivity (Mowshowitz and Kawaguchi, 2021).

### **3.3 Research Approach**

The study follows a deductive research methodology, in which the inferences are made based on the existing theories and empirical research. Deduction can be especially used in this research due to the fact that it is a continuation of the developed theoretical frameworks, such as the Technology Acceptance Model (TAM), Resource-Based View (RBV), and Stimulus-Organism-Response (S-O-R) model presented in Chapter 2.

The deductive method will allow the research to implement these theories in the setting of AI-driven SEO, and thus, challenge the appropriateness and explanatory value of these theories in the fast-paced digital world. As an illustration, TAM gives us information about the adoption of AI tools by marketers, whereas RBV elucidates the fact that AI capabilities can be used as strategic resources contributing to the competitive advantage (Assensoh-Kodua, 2019; Na et al., 2022). On the same note, the S-O-R model can be applied to understand the effect of AI-generated content on user behaviour and interaction (Biswas et al., 2025).

### **3.4 Research Design**

The research design used in this study is a qualitative research design, which utilises a secondary research approach (desk-based research). The qualitative design is suitable since the research aims to comprehend concepts, trends and strategic implications as opposed to quantifying numerical relationships.

The qualitative research approach will make it possible to critically and interpretively analyse the impact of AI technologies on SEO activities like content development, keyword search, and search engine positions. It allows one to combine both theoretical and practical views, thus closing the gap in the literature between the imaginary knowledge of concepts and their practical use (Cao and Tian, 2023). However, qualitative research has been highly criticised because it is subjective and cannot be generalised. To overcome this limitation in this study, the use of credible academic sources was used and system of approach to data analysis was adopted.

### **3.5 Data Sources**

The research is purely based on secondary data, comprising academic journal articles, books, industry reports and online databases like Google Scholar. Moreover, AI-based SEO tools (Semrush and Ahrefs) are also documented and analysed in practice to see how they are applied in practice.

Secondary data is also justified because the study will be based on synthesising existing knowledge and giving a holistic understanding of AI-based SEO. Secondary data is a way to access a significantly large number of studies and points of view, which will contribute to the depth and breadth of the analysis (Johnston, 2017). Moreover, since AI technologies are evolving quickly, secondary sources can give current information, which might be impossible to gather using primary research in a short period.

### **3.6 Data Collection Methods**

This research gathered information through a methodical literature review and case-based analysis of AI-based SEO tools. The literature review involved identifying, selecting, and critically analysing relevant academic and professional literature about AI and SEO. The systematic methodology will ensure that the data collection process is structured and transparent, and the risk of selection bias is reduced (Tranfield et al., 2003). In addition to the literature review, the analysis of the following tools, such as Semrush and Ahrefs, was performed with the help of the case-based analysis. Such tools will provide a practical insight into the application of AI in the sphere of SEO, including analysing keywords, competitor benchmarking and content optimisation. This will make the study viable as it bridges the theory-practical gap.

### **3.7 Analysis Techniques and justification of data**

They are used by the researcher as a primary approach to data analysis involving thematic analysis that adheres to the six-phase framework of data analysis proposed by Braun and Clarke (2006). It is a process, it involves the process of identifying, analysing and interpreting patterns in qualitative data. The thematic analysis particularly fits this study as it allows organising the complex data and arranging it into clear themes, such as AI-generated content, keyword intelligence, and algorithmic decision-making. Importantly, thematic analysis is customizable and adjustable, but it can also be biased by the author since it is subject to interpretation (Nowell et al., 2017). To address this shortcoming, the literature-based analysis in this paper is founded on the available literature and theoretical framework to make it consistent and reliable. Furthermore, the thematic analysis is also suitable in terms of the research purposes since it provides the possibility to determine the key trends and strategic implications of AI in SEO.

### **3.8 Ethical Considerations**

In this research paper, the acceptable ethics of research are adhered to. Being based on secondary data, it does not involve any human subjects, and therefore, such issues as informed consent and confidentiality are irrelevant. Ethical issues, however, are related to the aspect of academic integrity. Such sources have been referenced in order to eliminate plagiarism, and only sources that are credible and reliable have been used. In addition, the research also takes into consideration the ethical consequences of AI in SEO, including the problem of data privacy, algorithm bias, and the authenticity of content. These issues are critically analysed and discussed so that there is a balanced and responsible approach.

### **3.9 Research Limitations**

Although this study has strong methodological points, it has a number of limitations. The use of secondary data reduces the novelty of results and inhibits the possibilities of carrying out empirical validation. Moreover, the dynamism of AI technologies is such that certain findings might become outdated in the future.

The other weakness is that the study does not consider industry-specific analysis, given that the study follows a general approach to AI-driven SEO. This might not consider differences between sectors and organisational settings. Additionally, the application of qualitative analysis brings about some level of subjectivity since interpretations are subject to the viewpoint of the researcher.

### **3.10 Chapter Summary**

The chapter has also described the methodology used in carrying out the research with a critical reason behind the various methodological decisions taken. The research is based on a positivist philosophy and deductive approach, which is underpinned by a qualitative research design on the basis of secondary data. The systematic literature review and case-based analysis were employed to gather data and interpret the findings using a thematic analysis. In general, the selected methodology offers a solid base to examine how Artificial Intelligence has affected the practices of SEO.

## **4 Analysis and Discussion of Findings**

### **4.1 Introduction**

In this chapter, a thematic analysis and critical discussion of the findings based on a systematic literature review and case-based analysis performed in this study is presented. It is structured around five main themes discovered in the analysis process: (1) the evolution of SEO due to AI-based algorithms; (2) AI-based tools in content creation and key word intelligence; (3) AI-based SEO tools performance; (4) the usefulness of theoretical perspectives in understanding the concept of AI-driven SEO; and (5) the applicability of theoretical perspectives when it comes to understanding AI-driven SEO. The themes are critically discussed in relation to the existing literature, theoretical models and evidence of practical tools. This chapter ends with a synthesis to provide answers to the research questions and objectives stated in Chapter 1.

The thematic organisation adheres to the six-step model of Braun and Clarke (2006), making the analysis systematic, clear, and based on the data. Combining the theoretical views and the empirical evidence offered by the literature, this chapter offers a comprehensive understanding of how AI is changing the practice of SEO and what consequences it has on the business and professionals.

### **4.2 Theme 1: Transformation of SEO Through AI-Driven Algorithms**

The first is related to the radical change of SEO brought by AI-based search engine algorithms. This analysis indicates that search engines have experienced a paradigm shift of rule-based ranking to machine-learning-based analysis of relevance of content, user intent and contextual understanding. The most obvious example of such a change is the implementation of AI systems, like RankBrain (2015), BERT (2019), and MUM (2021), which are the next generations of natural language understanding (Elkhatibi and Boussedra, 2025).

The first deep-learning model implemented by Google was RankBrain, which processes search queries by mapping them to conceptually similar queries that Google has seen previously. According to Guelialia and Bouziane (2024), the introduction of RankBrain was a pivotal shift toward the strict matching of the keys to the meaning, towards the representation of the meaning within the vector space. Later, BERT (Bidirectional Encoder Representations from Transformers) greatly improved the capacity of Google to assess the entire context of words in a query as opposed to assessing words separately (Vashishth et al., 2025). This development underscored the significance of semantic content and writing in natural language rather than text that is full of keywords.

In 2021, Multitask Unified Model (MUM) added to this feature by allowing multimodal understanding, i.e. the processing of text, images, and video to provide more enriching search results (Sharma and Dhiman, 2025). All these algorithmic advances have made most of the old methods of SEO, including keyword stuffing and link farming, largely useless, forcing companies to shift to content strategies, focusing on depth, authority, and user value.

Theoretically, Information Asymmetry Theory is in line with these developments. The search engines have privileged access to large quantities of data, trends in user behaviour, and algorithmic designs, which practitioners lack, and there is an inherent imbalance of knowledge (Mowshowitz and Kawaguchi, 2021). Such asymmetry has serious strategic consequences: companies would have to invest in continuous monitoring, testing, and adjusting to remain visible in search results. This is further complicated by the fact that the ranking systems that are AI-driven are not transparent regarding the exact weight given to the factors.



In addition, the development of AI in search engines is consistent with the Resource-Based View (RBV), which assumes that organisations that can maintain a competitive edge need to create resources that are valuable, rare, and hard to copy (Kero and Bogale, 2023). When applied to the context of SEO, it suggests that those companies that can generate high-quality, semantically rich content that is backed by AI-informed strategies will have a sustainable competitive edge over those that use outdated optimisation strategies. Importantly, though, the RBV viewpoint should be deflated by the understanding that AI tools are open to everyone, i.e. the ownership of the tools is not the source of any advantage it is the strategic use of the tools that is advantageous (Hasan, 2025).

### **4.3 Theme 2: AI in Content Creation, Keyword Intelligence, and User Intent**

The second theme explores the use of AI in content creation, as well as, keyword research in the field of SEO. These two spheres are consistently reported in the literature as one of the most significantly impacted by the adoption of AI. The results indicate that AI has redirected the foundation of SEO content strategy to go beyond a single-keyword-focused strategy to a semantically holistic strategy based on the modelling of user intent.

Within the area of keyword intelligence, AI-based systems use natural language processing and machine learning to process large amounts of search data to find latent semantic associations between keywords and forecast future search trends. Sarfandi (2025) states that AI-enhanced keyword analysis allows marketers to identify so-called topic clusters: groups of semantically related keywords that indicate the variety of queries that users pose around a topic. This method is the opposite of old-fashioned keyword research, where the search volume and competition measurements were prioritised.

Also, AI enables the determination of the underlying intent of the search query, the reason the user was conducting a query, be it informational, navigational, transactional or commercial investigations. Search intent is becoming an increasingly important concept in the current state of SEO because Google algorithms are now more focused on the results that are the most relevant to the intent of a query (Ziakakis and Vlachopoulou, 2023). Artificial intelligence (AI) tools that can classify intent thus provide a major edge in content planning and optimisation.

The content production has also been changed by generative AI. Large language model (LLM)-based tools are capable of generating draft content, recommending structural changes, generating meta descriptions, and detecting content voids in comparison to competitors (Feuerriegel et al., 2024). Vajrobol et al. (2024) emphasise that generative AI has enabled organisations to scale the production of content on a scale previously unimaginable without commensurate increases in human resources expenditure.

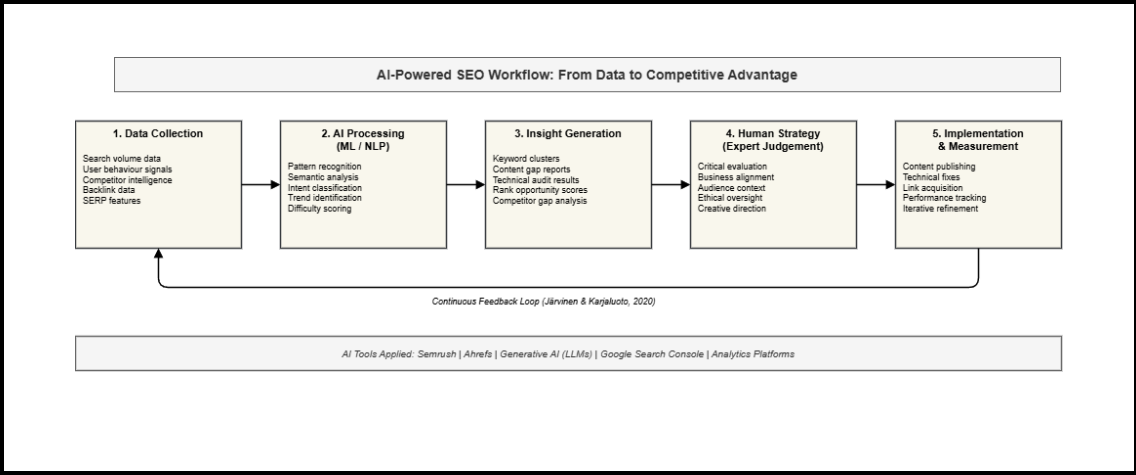
Nevertheless, the advantages of AI-generated content should be carefully considered. Cao and Tian (2023) warn that the proliferation of AI content tools is likely to be a source of homogenised output, which lacks originality and depth, and may weaken the differentiation that leads to organic search results. In the same manner, the search quality principles of Google still emphasise Expertise, Authoritativeness, and Trustworthiness (E-A-T) that attach the value of premium to the content that has been written by a knowledgeable human being (Sharma and Dhiman, 2025).

Here, the S-O-R framework (Biswas et al., 2025) is theoretically educative. AI-generated content is a stimulus; the cognitive and affective response by the user (organism) is the mediating factor on whether the content will result in the desired engagement behaviours (response), i.e., time-on-page, return visit, and conversions. This further confirms that AI content tools should be implemented not as standalone but as components of strategies based on profound knowledge of audience psychology and behaviour.

The table below outlines the essential changes in the practice of SEO that can be attributed to AI:

**Table 4.1: Shift in SEO Practice from Traditional to AI-Enhanced Approaches (adapted from Guelailia and Bouziane, 2024; Sarfandi, 2025)**

| SEO Dimension         | Traditional Approach            | AI-Enhanced Approach                          |
|-----------------------|---------------------------------|-----------------------------------------------|
| Keyword Research      | Volume/competition analysis     | Intent classification, topic clusters         |
| Content Creation      | Manual, keyword-centric writing | LLM-assisted, semantic depth                  |
| On-Page Optimisation  | Keyword density, meta tags      | Semantic relevance, E-A-T signals             |
| Link Building         | Quantity-focused outreach       | Authority-driven, AI-assessed relevance       |
| Performance Analytics | Page-level traffic metrics      | Predictive insights, user behaviour modelling |



**Figure 4.1: AI-Powered SEO Workflow - From Data Collection to Competitive Advantage (Our own, based on Järvinen & Karjaluoto, 2020; Sivahari, 2025)**

#### **4.4 Theme 3: Effectiveness of AI-Powered SEO Tools - Semrush and Ahrefs**

The third theme considers the feasibility of AI-powered search engine optimisation systems, and specifically Semrush and Ahrefs. These are the latest state-of-the-art in AI-assisted SEO, and they provide combined functions covering key intelligence, competitive analysis, backlink analysis, site audit and content optimisation (Sivahari, 2025). The use of machine learning enables Semrush to compute more than 20 billion keywords in various markets and present users with the difficulty of the keywords, the expected click-through rates, and trends. Its Keyword Magic Tool uses NLP to create semantic clusters to allow practitioners to create content architectures that encompass a topic holistically the approach favoured by Google to reward topical expertise (Chaffey and Ellis-Chadwick, 2022). The SEO Writing Assistant of Semrush also uses AI to help create the content by analysing the texts in real-time as to their readability, originality, tone, and semantic optimisation compared to the pages of the top competitors. Ahrefs, in its turn, uses mass data processing to utilise one of the largest backlink indices ever. Its Content Explorer leverages AI to explore billions of indexed pages, allowing users to find high-performing content in any niche, know what to share, and find link-building opportunities. According to Hasan (2025), the data-driven competitive intelligence offered by Ahrefs is a major development over manual benchmarking, which saves time taken in doing the entire competitive analysis to a few minutes. Industry evidence supports the effectiveness of these tools. Research mentioned by Sivahari (2025) suggests that those organisations that utilise AI-based SEO applications methodically report significant gains in organic search rankings, keyword rankings, and quality of traffic over those that use manual approaches. Ayoub and Sopuru (2026) also indicate that those companies that incorporate AI tools in restructured digital marketing processes have better performance results since the tools allow the dynamic capability, i.e. the capacity to sense, seize and reorganise strategic resources in relation to the strategic environment. Nevertheless, important restrictions should be taken into consideration. Semrush and Ahrefs are both commercial platforms that have proprietary datasets that do not comprehensively

capture the entirety of the web. There are different data currency levels, and the freshness of indexes influences the accuracy of recommendations (Sharma and Dhiman, 2025). Additionally, since the underlying algorithms are black-box, which implies that users have no full authority to audit the foundations on which they are getting the recommendations, there is a degree of epistemic dependency (Shafik 2026). Another issue is the over-reliance risk. By delegating the entire task of strategic decision-making in the field of strategic SEO to algorithmic recommendations without subjecting it to any critical human consideration, there is a high likelihood of organisations adhering to mechanically sound strategies of optimisation that are unlikely to be any strategic consideration, considering their position in the market, target market, or even the strategic objective. This issue supports the TAM-based understanding that the usefulness of AI tools depends on the ability of the user (Na et al., 2022).

#### **4.5 Theme 4: Benefits and Challenges of AI Integration in SEO**

The fourth theme is a synthesis of the advantages and the difficulties of the AI integration of SEO, which gives a balanced evaluation of the issue, in line with a critical approach taken in the present study

#### **4.5.1 Benefits**

The resources and time required to execute comprehensive SEO programs using AI tools are reduced many times by automation of labour-intensive processes, including the development of keywords, content audits, rank monitoring or competitor analysis (Jarvinen and Karjaluoto, 2020). This efficiency dividend is of particular interest to small organisations with small marketing budgets, where it can more easily compete with larger organisations on the basis of strategic intelligence, rather than resource base. The SEO insights are also more accurate and precise with the help of AI. Machine learning can identify trends of performance with the knowledge of datasets that previously were not accessible to human analysis mechanisms on a scale hitherto hidden, revealing insights which would be latent (Reddy et al., 2021). Predictive analytics has the ability to assist marketers in foreseeing ranking shifts, novel keyword trends, and competitor conduct, strategise based on proactivity instead of reactivity, and is accessible on websites like Semrush. Another benefit is personalisation, where AI can be used to provide personalised content experiences that appeal to certain user preferences and behavioural patterns (Kietzmann et al., 2023). This flexibility at the user level assists in pushing higher user interaction benefits, e.g. time on-site, bounce rate, frequency of returning, which is additionally applied as a measure that the search engine algorithm could take into account when determining the quality of the content.

### 4.5.2 Challenges

The most problematic consideration of AI in SEO is transparency. By design, as Mowshowitz and Kawaguchi (2021) note, AI-based ranking mechanisms are opaque, and search engines do not tell much about the precise criteria that they use to rank results. This ambiguity presents a strategic ambiguity since the practitioners will need to make optimisation decisions without being fully informed of the criteria on which their content will be evaluated. The same thing applies to algorithmic bias. The AI systems, which are trained on past search results, might reproduce or amplify the existing prejudices in the content perception and rankings, and discriminate against certain content, themes, or creators (Elkhatibi and Boussedra, 2025). This raises ethical concerns of equity, diversity, and structural consequences of the mediation of information through AI. The other problem is that of data privacy. Individualisation through AI and user intent modelling is founded on the abundance of data collected regarding the behaviour of users. Increasingly stringent data protection regulations, e.g. the General Data Protection Regulation (GDPR) in Europe, limit the quantity of user data that can be collected and processed, curtailing the capacity of AI tools to be tailored, and organisations worry about maximising objectives and adherence needs (Vajrobol et al., 2024). Finally, the technological change is an ongoing challenge. The search and SEO AI space is rapidly developing, and its unceasing progress requires being accompanied by investment in education, utilisation of tools, and re-strategising (Challoumis, 2024). Any of such organizations that fails to follow this transformation may find itself gradually waning in its organic search results and competitive advantages.

**Table 4.2: Benefits and Challenges of AI Integration in SEO**

| Factor     | Benefit / Challenge | Implication for Practice                          |
|------------|---------------------|---------------------------------------------------|
| Efficiency | Benefit             | Reduced manual effort; faster analysis cycles     |
| Accuracy   | Benefit             | Pattern detection at scale; data-driven decisions |

| Factor           | Benefit / Challenge | Implication for Practice                             |
|------------------|---------------------|------------------------------------------------------|
| Personalisation  | Benefit             | Improved engagement and conversion rates             |
| Transparency     | Challenge           | Strategic uncertainty in optimisation decisions      |
| Algorithmic Bias | Challenge           | Ethical and fairness concerns in content visibility  |
| Data Privacy     | Challenge           | Regulatory compliance constraints on personalisation |
| Pace of Change   | Challenge           | Continuous capability development required           |

#### 4.6 Theme 5: Theoretical Frameworks and Their Explanatory Power

The fifth theme analyses the explanatory strength of the three theoretical models presented in Chapter 2 to the empirical results of this research. The Technology Acceptance Model (TAM) is quite relevant when elucidating organisational attitudes towards AI-powered SEO tools. The results validate the idea that perceived usefulness is one of the key factors that prompt the use of tools, namely, the evident efficiency and insights that the use of tools like Semrush and Ahrefs can produce. The convenience factor is also important because both platforms have researched heavily into the interface design, guided workflows, and onboarding support (Na et al., 2022). The scope of explanation of TAM is, however, limited since it fails to explain the post-adoption challenges of strategic integration such as the misalignment in tool recommendation and organizational goals as observed in Theme 3. The Resource-Based View (RBV) can be helpful in understanding competitive aspect of AI-driven SEO. The analysis establishes that AI capabilities, when properly applied in the process of doing things in an organization, can be a source of competitive advantage. Nonetheless, the fundamental assumption of the RBV, that advantage is based on resources that are valuable, rare, and inimitable, is getting undermined in the context of commercially available AI tools, which can be purchased by any organization that accepts to pay subscription fees (Zvarimwa and Zimuto, 2022). This implies that



competitive advantage in AI-driven SEO lies not in tool access per se, but in organizational capabilities needed to use tool-generated insights in a strategic manner. The Stimulus-Organism-Response (S-O-R) model is an effective model that can be used to research how users react to AI-optimized content. The observation that AI-generated content can decrease differentiation (Cao and Tian, 2023) can be attributed to S-O-R predictions as when numerous stimuli (AI-generated content provided by rival organizations) are structurally similar, the organism will be less able to distinguish them, which lowers the chances of generating a strong behavioral response. This highlights the role played by human creativity in content AI-assisted concepts. Together, the three frameworks offer a complementary explanatory perspective: TAM is used to explain the behavior of adoption, RBV focuses on the strategic positioning, and S-O-R explains the outcomes of user engagement. The combination of them gives a more holistic analytical framework than any particular theory would give credit to, validating the usefulness of the multi-theoretical approach that is gained in this research.

#### **4.7 Discussion: Addressing the Research Questions**

**Research Question 1:** What has been the role of Artificial Intelligence in changing the search engine optimisation practices? This analysis shows that AI has radically re-oriented SEO as a technique of technical manipulation to a strategically advanced field. The application of machine learning algorithms by search engines, especially Google RankBrain, BERT, and MUM, has increased the significance of the quality of semantic content, the alignment of user intent, and overall topical authority in comparison to discrete keyword optimisation (Guelialia and Bouziane, 2024; Vashishth et al., 2025). Such a change demands business to implement integrated content approaches with the help of constant data analysis and optimisation with the aid of AI.

**Research Question 2:** How do AI technologies influence content creation, keyword searches, and results of search engines? The AI has also transformed the volume-based analysis of keywords to intent classification and topic cluster modelling (Sarfandi, 2025). Generative AI is a tool to scale production in content creation, but it poses the risk of homogenization and the dilution of quality (Feuerriegel et al., 2024; Cao and Tian, 2023). Semantic relevance, E-A-T indicators, and user interaction metrics are becoming the primary factors in search ranking, which AI can be used to optimise when implemented in a strategic manner.

**Research Question 3:** What is the effectiveness of AI-based SEO tools to enhance the performance of digital marketing? Semrush and Ahrefs tools have proven efficient, precise, and analytical capacity to boost the efficiency, accuracy, and depth of SEO programs (Sivahari, 2025; Hasan, 2025). But their usefulness is contingent upon the skill of the user and strategic incorporation, and they are subject to internal constraints of data completeness, the opacities of algorithms, and the over-reliance risk (Shafik, 2026; Ayoub and Soparu, 2026).

**Research Question 4:** What are the key challenges and constraints of AI in SEO? The issues of algorithmic privacy, algorithmic bias, and data privacy norms and changes in the rate are the most important. They are overlaid on the risk of becoming epistemically addicted to automated tools and the ethical implications of AI systems that create bias (Mowshowitz and Kawaguchi, 2021; Vajrobol et al., 2024; Elkhatibi and Boussedra, 2025).

**Research Question 5:** How can this be addressed to put businesses in a position to effectively adopt AI-based SEO practices? In order to achieve a successful implementation of AI, there should be a balance between AI-enhanced efficiency and human strategic decisions. The businesses should invest in data literacy, AI tool skills, and lifelong learning skills. Algorithms should not be the foundation of strategies but rather the interpretation of the audience and positioning of the brand. Ethically, AI-driven SEO governance systems must be realised in terms of such considerations as adherence to data privacy and bias monitoring (Challoubis, 2024; Ayoub and Sopuru, 2026).

## 4.8 Chapter Summary

The chapter is a thematic analysis of the findings, which is critical, and organised with five key themes, including algorithmic transformation of search engines, AI in content and keyword intelligence, tool effectiveness, benefits and challenges, and assessment of the theoretical frameworks. The answer to each of the five research questions was provided in the analysis and demonstrated that AI has brought a radical shift to the sphere of SEO, giving significant opportunities to be efficient and strategic and, on the one hand, being challenged by issues related to transparency, bias, and ethical governance. The findings are a strong empirical and theoretical foundation to the recommendations in Chapter 5.

## **5 Conclusions and Recommendations**

### **5.1 Introduction**

The chapter is the closing part of the thesis as it summarises the main findings of the study, explains their theoretical and practical implications and offers practical recommendations that businesses can implement, and research directions in the future. The chapter is organised in a way that it answers the research aim and objectives set in Chapter 1 and builds on the thematic analysis in Chapter 4. In line with the positivist and deductive methodological approach taken in the entire study, conclusions are made based on evidence as seen in the systematic literature review and case-based analysis.

### **5.2 Summary of Key Findings**

The thesis aimed at exploring the ways in which Artificial Intelligence has changed the way search engines are optimised, as well as to locate successful methods of applying AI to the practice of search engine optimisation. The enquiry led to the discovery of five major results: To begin with, AI has radically changed the operational and strategic basics of SEO. The introduction of AI-based algorithms such as Google RankBrain, BERT, and MUM has changed the foundation of search rankings to the quality of semantic content, alignment with user intent, and topical authority. This change makes the conventional method of SEO more and more outdated and requires a repositioning of the strategy towards AI-based content and optimisation (Guelialia and Bouziane, 2024; Vashishth et al., 2025). Second, AI has had an extensive impact on content creation and key intelligence. AI capabilities make it possible to classify intents, cluster topics, and produce content on a mass scale, thus increasing the efficiency and strategic soundness of SEO programs. Nevertheless, the dangers of the homogenization of content and its dilution related to the use of generative AI require close consideration alongside the human element of the editorial process (Feuerriegel et al., 2024; Cao and Tian, 2023). Third, AI-based SEO applications like Semrush and Ahrefs have proven to increase the level of

analytical depth, efficiency, and competitive intelligence practitioners have available. They can be effective, though, under the condition of user knowledge, strategy, and being aware of their inherent drawbacks, such as incomplete data and algorithmic non-transparency (Sivahari, 2025; Shafik, 2026). Fourth, although AI integration in SEO can be of great benefit, such as efficiency gains, increased accuracy, and the ability to offer personalisation, it can also present considerable problems. They include algorithmic bias, algorithmic privacy, data privacy compliance regulation, and the accelerated change in technology (Mowshowitz and Kawaguchi, 2021; Vajrobol et al., 2024; Challoumis, 2024). Fifth, the multi-theoretical framework of TAM, RBV, and S-O-R provides a deeper understanding of AI use, strategic location, and involvement in AI-based SEO than any other theoretical framework could allow. This not only builds theoretical knowledge in the field but also offers a framework of analysis that can be generalised in the future.

### **5.3 Theoretical Contributions**

The research offers a number of contributions to the scholarly literature on AI and digital marketing. First, it promotes the implementation of multi-theoretical frameworks in the AI-based research of SEO, which is indicated as a theoretically insufficient area in the literature review. Having shown the explanatory power of TAM, RBV and S-O-R to be complementary in the context, the study offers a replicable theoretical architecture to future scholarship. Second, the research provides a critical and balanced evaluation of the role of AI in SEO that combats the mostly positive focus of much of the current literature. The study contributes to the academic discussion on the intricate and multifaceted implications of AI adoption in marketing practice by empirically determining and analysing the limitations and risks of AI integration into marketing practice (Elkhatibi and Boussedra, 2025). Third, the research synthesises a disjointed literature and presents a unified analysis of AI-driven SEO, and it fills the research gap presented in Chapter 2. The thematic analytical framework created in Chapter 4 provides a systematic way of comprehending the effects of AI on SEO that could be used to guide future qualitative and empirical studies.

## 5.4 Practical Recommendations

Following the findings and discussion made in this thesis, the following practical recommendations are put forward to businesses, digital marketers and SEO professionals:

### **Recommendation 1: Implement an AI-Human Content Strategy.**

Companies ought to use generative AI-based solutions to hasten the ideation, writing, and optimisation of content, but keep human editorial supervision over content's substantive and alignment with E-A-T cues. Instead of replacing human creativity, AI ought to act as a catalyst for it (Feuerriegel et al., 2024; Sharma and Dhiman, 2025). Algorithms that review AI-generated content to determine its quality and suitability should be put in place to ensure that material is reviewed before publication.

### **Recommendation 2: Topical Authority and Semantic Content Architecture Investment.**

The new content strategies of businesses should be organised in terms of topic clusters and pillar pages, which capture the entire semantic environment of the field. The strategy is consistent with the topical authority and semantic depth preferred by the algorithm of Google, which is validated by the conclusions in Theme 2 (Sarfandi, 2025; Vashishth et al., 2025). Topic clusters can be systematically identified and mapped with the help of AI-driven tools like Semrush Keyword Magic Tool and Ahrefs Content Explorer.

### **Recommendation 3: Establish AI Literacy and Ability throughout the Organisation.**

The efficiency of AI-based SEO tools depends on the expertise of users (Na et al., 2022; Ayoub and Sopuru, 2026). Companies need to invest in formal training sessions to build AI literacy, data interpretation, and critical assessment skills among marketing and SEO practitioners. This is a consistent investment in RBV

principles since organisational capability is one of the factors determining the strategic value that is obtained through AI adoption, not access to tools.

**Recommendation 4: Adopt an Ethical AI Governance Framework.**

Organisations ought to develop governance solutions to oversee and reduce the risks of AI-driven SEO, which are algorithmic bias, data privacy compliance and content authenticity. This involves periodically auditing AI-driven content and tool-generated suggestions for potential bias or compliance problems. Related data collection and processing practices are to be in line with relevant data protection laws, such as GDPR (Vajrobol et al., 2024; Elkhatibi and Boussedra, 2025).

**Recommendation 5: Keep Adaptive Strategy and Continuous Monitoring.**

Since AI in search is evolving at an extremely fast rate, continuous environmental scanning and strategy revision need to be institutionalised in the organisation. SEO strategies must be checked and revised regularly--at least once a quarter--to consider algorithm changes and competitive changes. In order to stay competitive, continuous learning and staying up to date with new capabilities of AI are essential (Challoumis, 2024; Dwivedi et al., 2021).

## **5.5 Limitations of the Study**

There are limitations of this research that can be interpreted to explain the results. Firstly, due to the use of secondary data, there is not much empirical support for the conclusions. The absence of primary data, such as interview data from SEO professionals, and even longitudinal performance metrics for the businesses that have implemented AI tools, limits the research in terms of substantiating that the theoretical findings can be applied and implemented into the real world. Secondly, the dynamism of AI technologies means that some of the results and observations in the tool-specific sections may change as new functions are added to the tools. The analysis reflects the state of the art of the AI-based SEO practice, taken from the literature as of the time of writing. Third, the paper takes a broad approach to AI-based SEO and does not focus on industry-specific differences. The effects of AI on SEO could vary significantly depending on the industry, with varying competitive frameworks, the type of content, and patterns of user behaviour. Fourth, the qualitative analysis nature presents interpretive subjectivity. Although this has been alleviated by the systematic literature review processes and theoretical basis, one cannot be assured total objectivity in qualitative research (Nowell et al., 2017).



## 5.6 Directions for Future Research

The results and shortcomings of this research point to some fruitful avenues of research in the future. To begin with, empirical research, which utilises primary data collection methods, such as longitudinal case-studies, surveys of SEO professionals, and controlled experiments, would be an excellent source of evidence of the actual efficacy of AI-powered SEO tools in various organisational settings. Second, a study focusing on AI-driven SEO implementation and experience in specific industries, including healthcare, financial services, and e-commerce, would produce more specific and practical information compared to the general approach taken in the study. Third, studies which explore the ethical aspect of AI in SEO, especially algorithmic bias, content authenticity, and the potential effects of AI-mediated access to information on popular knowledge and democratic discourse, are a significant and under-researched area (Elkhatibi and Boussedra, 2025; Shafik, 2026). Fourth, further research could expand the multi-theoretical framework created in the current research by adding other perspectives, e.g., Dynamic Capabilities Theory (Ayoub and Sopuru, 2026) or institutional theory, to better explain the issue of AI adoption and strategic adaptation in the field of SEO. Lastly, in an age where generative AI functionality continues to evolve, studies that explore the future consequences of AI-generated content on search engine ecosystems, the quality of content, and user trust are both timely and important (Feuerriegel et al., 2024; Sharma and Dhiman, 2025).

## **5.7 Concluding Statement**

Information searching has been transformed by the application of Artificial Intelligence and has turned into more of a technologically manipulative research area into a data-driven, strategically robotic, user-value-centred, and high-quality semantic content practice. This thesis shows that AI has the potential to provide enormous opportunities to businesses willing to make investments in the capabilities, governance structures and dynamic plans to leverage AI potential. Meanwhile, the paper also indicates that the implementation of AI in the sphere of SEO is not entirely a secure issue: since the algorithms lack clarity, there is bias and homogenization of the generative AI, and critical consciousness and additional ethical reasoning are still required. The businesses are aware that AI cannot be passively adopted. The strategic value of the AI age of search is that organisations will be able to tap into the power and reasoning capacity of AI-based tools by getting the creativity, professionalism of the field and the ethical leadership that can only be offered by human practitioners who will be well informed. This paper adds a critically balanced and practical mind-based knowledge and walks in this changing terrain that has its basis in the theories.

## **Statement of AI Use**

In the preparation of this thesis, we used ChatGPT for generating structural headings for the literature review and analysis chapters, as well as revising the wording of certain paragraphs in the abstract and methodology sections, and for grammatical corrections throughout the thesis. We have reviewed and edited all the content after using this tool for factual accuracy, logical consistency and proper citation of sources. We have complete responsibility for the final text. AI-generated content without substantial human editing has not been used. No outputs of the AI were referenced as a source, nor were any outputs beyond their verification included in the thesis.

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